Q1 - 2024 (04 Apr Shift 1)

Let the first three terms $2, p$ and q, with $q \neq 2$, of a G.P. be respectively the 7^{th}, 8^{th} and 13^{th} terms of an A.P. If the 5^{th} term of the G.P. is the n^{th} term of the A.P., then n is equal to:

(1) 163

(2) 151

(3) 177

(4) 169

Q2 - 2024 (04 Apr Shift 2)

The value of $\frac{1\times2^2+2\times3^2+\ldots+100\times(101)^2}{1^2\times2+2^2\times3+\ldots.+100^2\times101}$ is

(1) $\frac{32}{31}$

(2) $\frac{31}{30}$

(3) $\frac{306}{305}$

(4) $\frac{305}{301}$

Q3 - 2024 (04 Apr Shift 2)

Let three real numbers a, b, c be in arithmetic progression and $a+1, b, c+3$ be in geometric progression. If $a > 10$ and the arithmetic mean of a, b and c is 8, then the cube of the geometric mean of a, b and c is

(1) 128

(2) 316

(3) 120

(4) 312

Q4 - 2024 (05 Apr Shift 1)

If $\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\ldots+\frac{1}{\sqrt{99}+\sqrt{100}}=m$ and $\frac{1}{1\cdot2}+\frac{1}{2\cdot3}+\ldots+\frac{1}{99\cdot100}=n$, then the point (m, n) lies on the line

(1) $11(x-1)-100(y-2)=0$

(2) $11x - 100y = 0$

(3) $11(x-2) - 100(y-1) = 0$

(4) $11(x-1) - 100y = 0$

Q5 - 2024 (05 Apr Shift 1)

Let $a_1, a_2, a_3, \ldots$ be in an arithmetic progression of positive terms.

Let $\mathrm{A_k = a_1^2 - a_2^2 + a_3^2 - a_4^2 + \ldots + a_{2k-1}^2 - a_{2k}^2}$.

If $\mathrm{A_3 = -153,\ A_5 = -435}$ and $\mathrm{a_1^2 + a_2^2 + a_3^2 = 66}$, then $\mathrm{a_{17} - A_7}$ is equal to______

Q6 - 2024 (05 Apr Shift 2)

For $x \geqslant 0$, the least value of K, for which $4^{1+x} + 4^{1-x}, \frac{\mathrm{K}}{2}, 16^x + 16^{-x}$ are three consecutive terms of an A.P., is equal to :

(1) 8

(2) 4

(3) 10

(4) 16

Q7 - 2024 (05 Apr Shift 2)

If $1 + \frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}} + \frac{5-2\sqrt{6}}{18} + \frac{9\sqrt{3}-11\sqrt{2}}{36\sqrt{3}} + \frac{49-20\sqrt{6}}{180} + \ldots \text{ upto } \infty = 2 + \left(\sqrt{\frac{b}{a}} + 1\right) \log_e\left(\frac{a}{b}\right)$, where a and b are integers with $\mathrm{gcd(a, b) = 1}$, then $\mathrm{11a + 18\,b}$ is equal to ______

Q8 - 2024 (06 Apr Shift 1)

Let $A = \{n \in [100, 700] \cap \mathbb{N} : n \text{ is neither a multiple of 3 nor a multiple of 4}\}$. Then the number of elements in A is

(1) 290

(2) 280

(3) 300

(4) 310

Q9 - 2024 (06 Apr Shift 1)

Let the first term of a series be $T_1 = 6$ and its r^{th} term $T_r = 3T_{r-1} + 6^r, r = 2, 3,$

n. If the sum of the first n terms of this series is $\frac{1}{5}\left(n^2 - 12n + 39\right)\left(4 \cdot 6^n - 5 \cdot 3^n + 1\right)$, then n is equal to______

Q10 - 2024 (06 Apr Shift 2)

Let ABC be an equilateral triangle. A new triangle is formed by joining the middle points of all sides of the triangle ABC and the same process is repeated infinitely many times. If P is the sum of perimeters and Q is be the sum of areas of all the triangles formed in this process, then :

(1) $\mathrm{P}^2 = 6\sqrt{3}\mathrm{Q}$

(2) $\mathrm{P}^2 = 36\sqrt{3}\mathrm{Q}$

(3) $\mathrm{P} = 36\sqrt{3}\mathrm{Q}^2$

(4) $\mathrm{P}^2 = 72\sqrt{3}\mathrm{Q}$

Q11 - 2024 (06 Apr Shift 2)

A software company sets up m number of computer systems to finish an assignment in 17 days. If 4 computer systems crashed on the start of the second day, 4 more computer systems crashed on the start of the third day and so on, then it took 8 more days to finish the assignment. The value of m is equal to:

(1) 150

(2) 180

(3) 160

(4) 125

Q12 - 2024 (06 Apr Shift 2)

If $\mathrm{S}(x)=(1+x)+2(1+x)^2+3(1+x)^3+\cdots+60(1+x)^{60}, x \neq 0$, and $(60)^2 \mathrm{~S}(60)=\mathrm{a}(\mathrm{b})^{\mathrm{b}}+\mathrm{b}$, where $a, b \in N$, then $(a+b)$ equal to ______

Q13 - 2024 (08 Apr Shift 1)

If the set $R=\{(a, b): a+5 b=42, a, b \in \mathbb{N}\}$ has m elements and $\sum_{n=1}^m\left(1-i^{n!}\right)=x+i y$, where $i=\sqrt{-1}$, then the value of $m+x+y$ is

(1) 12

(2) 4

(3) 8

(4) 5

Q14 - 2024 (08 Apr Shift 1)

Let the positive integers be written in the form :

1

2 3

4 5 6

7 8 9 10

. . . . . .

If the k^{th} row contains exactly k numbers for every natural number k, then the row in which the number 5310 will be, is ______

Q15 - 2024 (08 Apr Shift 2)

In an increasing geometric progression of positive terms, the sum of the second and sixth terms is $\frac{70}{3}$ and the product of the third and fifth terms is 49 . Then the sum of the 4^{th}, 6^{th} and 8^{th} terms is equal to :

(1) 96

(2) 91

(3) 84

(4) 78

Q16 - 2024 (08 Apr Shift 2)

An arithmetic progression is written in the following way

						2							
				5				8					
		11				14				17			
20				23				26				29	

$- -$

$- -$

The sum of all the terms of the 10^{th} row is________

Q17 - 2024 (09 Apr Shift 1)

If the sum of the series $\frac{1}{1\cdot(1+d)}+\frac{1}{(1+d)(1+2\,d)}+\ldots+\frac{1}{(1+9\,d)(1+10\,d)}$ is equal to 5 , then 50 d is equal to :

(1) 10

(2) 5

(3) 15

(4) 20

Q18 - 2024 (09 Apr Shift 2)

Let $a, ar, ar^2,$ be an infinite G.P. If $\sum_{n=0}^{\infty} ar^n = 57$ and $\sum_{n=0}^{\infty} a^3r^{3n} = 9747$, then $a + 18r$ is equal to

(1) 46

(2) 38

(3) 31

(4) 27

Q19 - 2024 (09 Apr Shift 2)

If $\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots\ldots+\frac{1}{\alpha+1012}\right)-\left(\frac{1}{2\cdot1}+\frac{1}{4\cdot3}+\frac{1}{6\cdot5}+\ldots\ldots+\frac{1}{2024\cdot2023}\right)=\frac{1}{2024}$, then α is equal to________

Answer Key

Q1 (1)	**Q2** (4)	**Q3** (3)	**Q4** (2)
Q5 (910)	**Q6** (3)	**Q7** (76)	**Q8** (3)
Q9 (6)	**Q10** (2)	**Q11** (1)	**Q12** (3660)
Q13 (1)	**Q14** (103)	**Q15** (2)	**Q16** (1505)
Q17 (2)	**Q18** (3)	**Q19** (1011)	

Q1

$p^2 = 2q$

$2 = a + 6d$ ...(i)

$\mathrm{p} = \mathrm{a} + 7\ \mathrm{d}$ ...(ii)

$q = a + 12d$ ...(iii)

$\mathrm{p} - 2 = \mathrm{d}$ ((ii) - (i))

$q - p = 5d$ ((iii) - (ii))

$q - p = 5d$
$q - p = 5(p - 2)$
$q = 6p - 10$
$p^2 = 2(6p - 10)$
$p^2 - 12p + 20 = 0$
$p = 10, 2$
$p = 10; q = 50$
$d = 8$
$a = -46$
$2, 10, 50, 250, 1250$
$a^4 = a + (n-1)d$
$1250 = -46 + (n-1)8$
$n = 163$

Q2

$$\frac{1 \times 2^2 + 2 \times 3^2 + \ldots + 100 \times (101)^2}{1^2 \times 2 + 2^2 \times 3 + \ldots + 100^2 \times 101} = \frac{\sum_{r=1}^{100} r(r+1)^2}{\sum_{r=1}^{100} r^2(r+1)}$$

$$= \frac{\sum_{r=1}^{100}\left(r^3 + 2r^2 + r\right)}{\sum_{r=1}^{100}\left(r^3 + r^2\right)} = \frac{\left(\frac{n(n+1)^2}{2}\right) + \frac{2 \cdot n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}}{\left(\frac{n(n+1)}{2}\right)^2 + \frac{n(n+1)(2n+1)}{6}}$$

$$= \frac{\frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} + \frac{2}{3} \cdot (2n+1) + 1\right]}{\frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} + \frac{(2n+1)}{3}\right]};\ \text{Put n=100}$$

$$= \frac{\frac{100(101)}{2} + \frac{2}{3}(201) + 1}{\frac{100 \times 101}{2} + \frac{201}{3}} = \frac{5185}{5117} = \frac{305}{301}$$

Q3

$2b = a + c, b^2 = (a+1)(c+3)$

$\dfrac{a+b+c}{3} = 8 \to b = 8, a + c = 16$

$64 = (a+1)(19-a) = 19 + 18a - a^2$

$a^2 - 18a - 45 = 0 \to (a-15)(a+3) = 0, (a > 10)$

$a = 15, c = 1, b = 8$

$\left((abc)^{1/3}\right)^3 = abc = 120$

Q4

$\dfrac{1}{\sqrt{1}+\sqrt{2}} + \dfrac{1}{\sqrt{2}+\sqrt{3}} + \ldots + \dfrac{1}{\sqrt{99}+\sqrt{100}} = \mathrm{m}$

$\dfrac{\sqrt{1}-\sqrt{2}}{-1} + \dfrac{\sqrt{2}-\sqrt{3}}{-1} \ldots \dfrac{\sqrt{99}-\sqrt{100}}{-1} = \mathrm{m}$

$\sqrt{100} - 1 = \mathrm{m} \Rightarrow \mathrm{m} = 9$

$\dfrac{1}{1 \cdot 2} + \dfrac{1}{2 \cdot 3} + \ldots \dfrac{1}{99 \cdot 100} = \mathrm{n}$

$\dfrac{1}{1} - \dfrac{1}{2} + \dfrac{1}{2} - \dfrac{1}{3} \ldots \dfrac{1}{99} - \dfrac{1}{100} = \mathrm{n}$

$1 - \dfrac{1}{100} = \mathrm{n}$

$\dfrac{99}{100} = \mathrm{n}$

$(\mathrm{m}, \mathrm{n}) = \left(9, \dfrac{99}{100}\right)$

$\Rightarrow 11(9) - 100\left(\dfrac{99}{100}\right)$

$= 99 - 99 = 0$

Ans. $11x - 100y = 0$

Q5

$d \rightarrow$ common diff.

$A_k = -kd[2a + (2k-1)d]$

$A_3 = -153$

$\Rightarrow 153 = 13\,d[2a + 5\,d]$

$51 = d[2a + 5\,d] \ldots (1)$

$A_5 = -435$

$435 = 5\,d[2a + 9\,d]$

$87 = d[2a + 9\,d]$

$(2) - (1)$

$36 = 4\,d^2$

$d = 3, a = 1$

$a_{17} - A_7 = 49 - [-7.3[2 + 39]] = 910$

Q6

$$k = 4\underbrace{\left(4^x + \frac{1}{4^x}\right)}_{\geq 2} + \underbrace{\left(4^{2x} + \frac{1}{4^{2x}}\right)}_{\geq 2}$$

$k \geq 10$

Q7

$$S=1+\frac{x}{2\sqrt{3}}+\frac{x^2}{18}+\frac{x^3}{36\sqrt{3}}+\frac{x^4}{180}+\ldots\infty$$

Put $\frac{x}{\sqrt{3}}=t$, where $x=\sqrt{3}-\sqrt{2}$

$$S=1+\frac{t}{2}+\frac{t^2}{6}+\frac{t^3}{12}+\frac{t^4}{20}+\ldots$$

$$S=1+t\left(1-\frac{1}{2}\right)+t^2\left(\frac{1}{2}-\frac{1}{3}\right)+t^3\left(\frac{1}{3}-\frac{1}{4}\right)+t^4\left(\frac{1}{4}-\frac{1}{5}\right)$$

$$S=\left(1+t+\frac{t^2}{2}+\frac{t^3}{3}+\frac{t^3}{4}+\ldots\right)-\left(\frac{t}{2}+\frac{t^2}{3}+\frac{t^3}{4}+\frac{t^4}{5}+\ldots\right)$$

$$S=\left(t+\frac{t^2}{2}+\ldots\right)-\frac{1}{t}\left(t+\frac{t^2}{2}+\frac{t^3}{3}+\ldots\right)+2$$

$$S=2+\left(1-\frac{1}{t}\right)(-\log(1-t))=\left(\frac{1}{t}-1\right)\log(1-t)+2$$

$$S=2+\left(\frac{\sqrt{3}}{\sqrt{3}-\sqrt{2}}-1\right)\log\left(1-\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}}\right)$$

$$S=2+\left(\frac{\sqrt{2}}{\sqrt{3}-\sqrt{2}}\right)\log e\frac{\sqrt{2}}{\sqrt{3}}$$

$$S=2+\frac{(\sqrt{6}+2)}{2}\log e\frac{2}{3}=2+\left(\sqrt{\frac{3}{2}}+1\right)\log e\frac{2}{3}$$

$a=2, b=3$

$11a+18b=11\times 2+18\times 3=76$

Q8

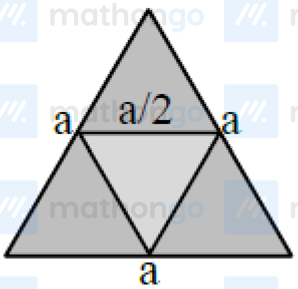
$\mathrm{n}(3) \Rightarrow$ multiple of 3
$102, 105, 108, \ldots\ldots, 699$
$\mathrm{T}_{\mathrm{n}} = 699 = 102 + (\mathrm{n}-1)(3)$
$\mathrm{n} = 200$
$\mathrm{n}(3) = 200$
$\because \mathrm{n}(4) \Rightarrow$ multiple of 4
$100, 104, 108, \ldots\ldots, 700$
$T_n = 700 = 100 + (n-1)(4)$
$n = 151$
$n(4) = 151$
$n(3 \cap 4) \Rightarrow$ multiple of 3&4 both
$108, 120, 132, \ldots\ldots, 696$
$T_n = 696 = 108 + (n-1)(12)$
$n = 50$
$n(3 \cap 4) = 50$
$n(3 \cup 4) = n(3) + n(4) - n(3 \cap 4)$
$= 200 + 151 - 50$
$= 301$
$\overline{n(3 \cup 4)} =$ Total $- n(3 \cup 4) =$ neither a multiple of 3 nor a multiple of 4
$= 601 - 301 = 300$

Q9

$\mathrm{T}_{\mathrm{r}} = 3\,\mathrm{T}_{\mathrm{r}-1} + 6^{\mathrm{r}}, \mathrm{r} = 2, 3, 4, \ldots \mathrm{n}$
$\mathrm{T}_2 = 3.\,\mathrm{T}_1 + 6^2$
$\mathrm{T}_2 = 3.6 + 6^2 \ldots (1)$
$\mathrm{T}_3 = 3\,\mathrm{T}_2 + 6^3$
$\mathrm{T}_3 = 3\,\mathrm{T}_2 + 6^3$
$\mathrm{T}_3 = 3\left(3.6 + 6^2\right) + 6^3$
$\mathrm{T}_3 = 3^2.6 + 3.6^2 + 6^3 \ldots (2)$

$$T_r = 3^{r-1} \cdot 6\left[1+\frac{6}{3}+\left(\frac{6}{3}\right)^2+\ldots+\left(\frac{6}{3}\right)^{r-1}\right]$$
$$T_r = 3^{r-1} \cdot 6\left(1+2+2^2+\ldots+2^{r-1}\right)$$
$$T_r = 6 \cdot 3^{r-1} 1 \cdot \frac{\left(1-2^r\right)}{(-1)}$$
$$T_r = 6 \cdot 3^{r-1} \cdot \left(2^r-1\right)$$
$$T_r = \frac{6 \cdot 3^r}{3} \cdot \left(2^r-1\right)$$
$$T_r = 2 \cdot \left(6^r-3^r\right)$$
$$S_n = 2\Sigma\left(6^r-3^r\right)$$
$$S_n = 2 \cdot\left[\frac{6.\left(6^n-1\right)}{5}-\frac{3.\left(3^n-1\right)}{2}\right]$$
$$S_n = 2\left[\frac{12\left(6^n-1\right)-15\left(3^n-1\right)}{10}\right]$$
$$S_n = \frac{3}{5}\left[4 \cdot 6^4-5 \cdot 3^n+1\right]$$
$$\therefore n^2-12n+39=3$$
$$n^2-12n+36=0$$
$$n=6$$

Q10



Area of first $\Delta = \frac{\sqrt{3}a^2}{4}$

Area of second $\Delta = \frac{\sqrt{3}a^2}{4} \frac{a^2}{4} = \frac{\sqrt{3}a^2}{16}$

Area of third $\Delta = \frac{\sqrt{3}a^2}{64}$

sum of area $= \frac{\sqrt{3}a^2}{4}\left(1+\frac{1}{4}+\frac{1}{16}\ldots\right)$

$$Q = \frac{\sqrt{3}a^2}{4}\frac{1}{\frac{3}{4}} = \frac{a^2}{\sqrt{3}}$$

perimeter of $1^{\text{st}}\ \Delta = 3a$

perimeter of $2^{\text{nd}}\ \Delta = \frac{3a}{2}$

perimeter of $3^{\text{rd}}\ \Delta = \frac{3a}{4}$

$$P = 3a\left(1+\frac{1}{2}+\frac{1}{4}+\ldots\right)$$

$$P = 3a \cdot 2 = 6a$$

$$a = \frac{P}{6}$$

$$Q = \frac{1}{\sqrt{3}} \cdot \frac{P^2}{36}$$

$$P^2 = 36\sqrt{3}Q$$

Q11

$$17\text{ m} = \text{m} + (\text{m}-4) + (\text{m}-4\times 2)\ldots + \ldots(\text{m}-4\times 24)$$

$$17\text{ m} = 25\text{ m} - 4(1+2\ldots 24)$$

$$8\text{ m} = \frac{4\cdot 24\cdot 25}{2} = 150$$

Q12

$$\begin{aligned} S(x) =& (1+x)+2(1+x)^2+3(1+x)^3+\ldots+60(1+x)^{60} \\ (1+x)S =& (1+x)^2+\ldots\ldots\quad 59(1+x)^{60}+60(1+x)^{61} \\ -xS =& \frac{(1+x)(1+x)^{60}-1}{x}-60(1+x)^{61} \end{aligned}$$

Put x = 60

$$-60\text{ S} = \frac{61((61)^{60}-1)}{60} - 60(61)^{61}$$

on solving 3660

Q13

$a+5b=42, a, b \in N$
$a=42-5b, b=1, a=37$
$b=2, a=32$
$b=3, a=27$
$\vdots$
$b=8, a=2$

R has " 8 " elements $\Rightarrow \mathrm{m}=8$

$$\sum_{n=1}^{8}\left(1-i^{n!}\right)=x+iy$$

for $n \geq 4, i^{n!}=1$

$\Rightarrow (1-i)+\left(1-i^{2!}\right)+\left(1-i^{3!}\right)$
$=1-I+2+1+1$
$=5-I=x+iy$
$m+x+y=8+5-1=12$

Q14

$\mathrm{S}=1+2+4+7+\ldots\ldots+\mathrm{T}_{\mathrm{n}}$
$\mathrm{S}=1+2+4+\ldots\ldots$
$\mathrm{Tn}=1+1+2+3+\ldots\ldots+(\mathrm{T}_{\mathrm{n}}-\mathrm{T}_{\mathrm{n}-1})$

$$\mathrm{T}_{\mathrm{n}}=1+\left(\frac{\mathrm{n}-1}{2}\right)[2+(\mathrm{n}-2)\times 1]$$

$$\mathrm{T}_{\mathrm{n}}=1+1+\frac{\mathrm{n}(\mathrm{n}-1)}{2}$$

$\mathrm{n}=100 \quad \mathrm{T}_{\mathrm{n}}=1+\dfrac{100\times 99}{2}=4950+1$

$\mathrm{n}=101 \quad \mathrm{T}_{\mathrm{n}}=1+\dfrac{101\times 100}{2}=5050+1=5051$

$\mathrm{n}=102 \quad \mathrm{T}_{\mathrm{n}}=1+\dfrac{102\times 101}{2}=5151+1=5152$

$\mathrm{n}=103 \quad \mathrm{T}_{\mathrm{n}}=1+\dfrac{103\times 102}{2}=5254$

$\mathrm{n}=104 \quad \mathrm{T}_{\mathrm{n}}=1+\dfrac{104\times 103}{2}=5357$

Q15

$\mathrm{T}_2+\mathrm{T}_6=\frac{70}{3}$
$\mathrm{ar}+\mathrm{ar}^5=\frac{70}{3}$
$\mathrm{T}_3\cdot\mathrm{T}_5=49$

$\mathrm{ar}^2 \cdot \mathrm{ar}^4 = 49$

$\mathrm{a}^2\mathrm{r}^6 = 49$

$\mathrm{ar}^3 = +7, \mathrm{a} = \frac{7}{\mathrm{r}^3}$

$\mathrm{ar}\left(1+\mathrm{r}^4\right) = \frac{70}{3}$

$\frac{7}{\mathrm{r}^2}\left(1+\mathrm{r}^4\right) = \frac{70}{3}, \mathrm{r}^2 = \mathrm{t}$

$\frac{1}{\mathrm{t}}\left(1+\mathrm{t}^2\right) = \frac{10}{3}$

$3\mathrm{t}^2 - 10\mathrm{t} + 3 = 0$

$\mathrm{t} = 3, \frac{1}{3}$

Increasing G.P.

$\mathrm{r}^2 = 3, \mathrm{r} = \sqrt{3}$

$\mathrm{T}_4 + \mathrm{T}_6 + \mathrm{T}_8$

$= \mathrm{ar}^3 + \mathrm{ar}^5 + \mathrm{ar}^7$

$= \mathrm{ar}^3\left(1+\mathrm{r}^2+\mathrm{r}^4\right)$

$= 7(1+3+9) = 91$

Q16

$2, 5, 11, 20, \ldots\ldots$

$$\text{General term } = \frac{3n^2 - 3n + 4}{2}$$

$$\mathrm{T}_{10} = \frac{3(100) - 3(10) + 4}{2}$$

$= 137$

10 terms with c.d. $= 3$

$$\text{sum } = \frac{10}{2}(2(137) + 9(3))$$

$= 1505$

Q17

$$\frac{1}{1\cdot(1+\mathrm{d})}+\frac{1}{(1+\mathrm{d})(1+2\ \mathrm{d})}+\ldots\ldots\ldots\ldots\ldots$$

$$\frac{1}{(1+9\ \mathrm{d})(1+10\ \mathrm{d})}=5$$

$$\frac{1}{d}\left[\frac{(1+d)-1}{1\cdot(1+d)}+\frac{(1+2d)-(1-d)}{(1+d)(1+2d)}\right]+\ldots\ldots\ldots\ldots\ldots$$

$$\frac{(1+10d)-(1+9d)}{(1+9d)(1+10d)}=5$$

$$\frac{1}{d}\left[\left(1-\frac{1}{1+d}\right)+\left(\frac{1}{1+d}-\frac{1}{1+2d}\right)+\ldots\ldots\ldots\ldots\ldots\right.$$

$$\left.\left(\frac{1}{1+9d}-\frac{1}{1+10d}\right)\right]=5$$

$$\frac{1}{d}\left[1-\frac{1}{(1+10d)}\right]=5$$

$$\frac{10d}{1+10d}=5d$$

$$50d=5$$

Q18

$$\sum_{n=0}^{\infty}\mathrm{ar}^{\mathrm{n}}=57$$

$$\mathrm{a}+\mathrm{ar}+\mathrm{ar}^2+\infty=57$$

$$\frac{\mathrm{a}}{1-\mathrm{r}}=57\ldots(I)$$

$$\sum_{\mathrm{n}=0}^{\infty}\mathrm{a}^3\mathrm{r}^{3\mathrm{n}}=9747$$

$$\mathrm{a}^3+\mathrm{a}^3\cdot\mathrm{r}^3+\mathrm{a}^3\cdot\mathrm{r}^6+\ldots\ldots\ldots\ldots\ldots=9746$$

$$\frac{\mathrm{a}^3}{1-\mathrm{r}^3}=9746\ldots(II)$$

$$\frac{(\mathrm{I})^3}{(\mathrm{II})}\Rightarrow\frac{\frac{\mathrm{a}^3}{(1-\mathrm{r})^3}}{\frac{\mathrm{a}^3}{1-\mathrm{r}^3}}=\frac{57^3}{9717}=19$$

$$\mathrm{a}=19$$

$$\therefore \mathrm{a}+18\mathrm{r}=19+18\times\frac{2}{3}=31$$

On solving, $\mathrm{r}=\frac{2}{3}$ and $\mathrm{r}=\frac{3}{2}$ (rejected)

$$a=19$$

$$\therefore a+18r=19+18\times\frac{2}{3}=31$$

Q19

$$\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}\right)$$
$$-\left\{\left(\frac{1}{1}-\frac{1}{2}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+\ldots+\left(\frac{1}{2023}-\frac{1}{2024}\right)\right\}=\frac{1}{2024}$$
$$\Rightarrow\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}\right)$$
$$-\left\{\left(\frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}\right)+\ldots+\frac{1}{2023}\right.$$
$$\left.-\frac{1}{2024}-2\left(\frac{1}{2}+\frac{1}{4}+\ldots+\frac{1}{2022}\right)\right\}=\frac{1}{2024}$$

$$\Rightarrow\left(\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}\right)$$
$$-\left(\frac{1}{1}+\frac{1}{2}+\ldots+\frac{1}{2023}\right)$$
$$+\frac{1}{2024}+\left(\frac{1}{1}+\frac{1}{2}+\ldots+\frac{1}{1011}\right)=\frac{1}{2024}$$
$$\Rightarrow\frac{1}{\alpha+1}+\frac{1}{\alpha+2}+\ldots+\frac{1}{\alpha+2012}$$
$$=\frac{1}{1012}+\frac{1}{1013}+\ldots+\frac{1}{2023}$$
$$\Rightarrow\alpha=1011$$